



OPEN Improving the resilience of urban transportation to natural disasters: the case of Changchun, China

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This study quantitatively assesses the resilience of the urban transport system in Changchun under extreme climatic conditions, focusing on the impacts of natural disasters such as snowstorms, strong winds and extreme low temperatures on the transport system. The vulnerability, exposure, and emergency recovery capacity of the transport system in Changchun were analyzed by constructing a comprehensive assessment framework combining multi-criteria decision analysis (MCDM) and geographic information system (GIS). Based on the meteorological and traffic data of Changchun City in the past 10 years, key indicators such as traffic network density, emergency resource distribution, traffic flow, and extreme weather frequency were selected in this study. The weights were calculated by the improved hierarchical analysis method (AHP) and entropy weighting method, and finally the traffic resilience of Changchun city was quantitatively analyzed. The results show that there are significant spatial differences in the toughness of the transport system in Changchun, of which 38.6% of the areas have very low toughness and are concentrated in the urban fringe and urban–rural areas; 25.3% of the areas are at a medium level of toughness; and 5.4% of the areas show high toughness and are mainly located in the city center and important transport hubs. This study proposes transport resilience enhancement strategies for different areas of Changchun, with special recommendations for strengthening infrastructure development, optimizing traffic management and emergency response capacity in low resilience areas. The results of this study provide a scientific basis for Changchun to cope with future extreme climate challenges, and a reference framework for other cities with similar climatic conditions.

Keywords Natural hazard, Accurate resilience evaluation, Remote sensing and GIS, Multi-criteria decision analysis

With the intensification of global climate change, the impact of extreme weather events on urban transport systems is becoming increasingly serious, especially in areas with complex climates and frequent extreme weather, and the resilience of transport systems has become a key issue in urban management^{1,2}. Transport resilience not only refers to the impact resistance of the transport system in disasters, but also pays more attention to its ability to recover rapidly after disasters^{3,4}. In recent years, scholars at home and abroad have carried out extensive research on the resilience of transport systems from different perspectives, but there are still some limitations in the research on coping with the extreme climatic conditions in Northeast China⁵.

The concept of transport resilience is derived from infrastructure resilience studies and usually covers the vulnerability, adaptability and resilience of a system⁶. Transport resilience not only involves the strength of physical infrastructure (e.g., roads and bridges), but also the system's flexibility and ability to adjust in a disaster^{7,8}. Wells et al.⁹ proposed a framework for analyzing the resilience of infrastructure, which takes into account the impacts of multiple factors on the transport system, and provides a quantitative assessment tool for city managers. Extreme weather such as blizzards, low temperatures and strong winds pose a significant threat to the transport system in Northeast Asia¹⁰. Zhao et al.¹¹ investigated the impact of blizzards on the transport system in Northeast China, pointing out that extreme low temperatures and heavy snowfalls not only cause damage to roads, but also lead to an increase in the rate of traffic accidents. Yin and Sun¹² explored the impact of China's blizzard disaster in Northeast China, highlighting the importance of vulnerability and

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post-disaster resilience of transport systems. Extreme weather not only affects physical facilities, but also brings disruptions and delays to the functional operations of transport systems¹³. For example, Patrascu et al.¹⁴ studied how extreme weather affects traffic disruptions in Texas, USA, showing that low-lying areas and transport hubs are more vulnerable in the face of extreme weather. This study provides important insights into understanding the regional vulnerability of urban transport systems¹⁵. In recent years, resilience assessment frameworks based on multi-criteria decision analysis (MCDM) and geographic information system (GIS) technologies have been widely used^{16,17}. The MCDM framework proposed by Tehler et al.¹⁸ can integrate multiple resilience assessment indicators, quantify the transport resilience of different regions, and provide a scientific basis for regional planning. Meanwhile, GIS technology reveals the differences in system vulnerability in different regions through spatial analysis of the transport network, especially in the context of extreme weather such as snowstorms, GIS plays a crucial role in identifying areas of high vulnerability¹⁹.

In recent years, research on the resilience of urban transportation systems has gradually received widespread attention, and most scholars have explored it from the perspectives of infrastructure strength, disaster recovery capability and system adaptability. However, the existing studies still have the following shortcomings: first, most of the studies focus on the assessment of single-hazard scenarios and lack a systematic analysis of multi-hazard scenarios under complex climatic conditions. For example, Lu et al.²⁰ study only explores the impacts of heavy rainfall on the transportation system, while ignoring the comprehensive impacts on the system when multiple weather extremes (e.g., strong winds, low temperatures, and short-duration heavy rainfall) occur alternately or in combination. This one-sidedness limits the applicability of resilience assessment results in practical disaster management. Second, the existing assessment frameworks mostly focus on the vulnerability analysis at the physical level, while the research on the functional level of the transportation system (e.g., emergency response capacity and recovery capacity) is relatively insufficient²¹. However, functional recovery of transportation systems plays a crucial role in post-disaster emergency response and long-term resilience building. Neglecting this dimension may lead to limited effectiveness of enhancement strategies. Although multi-criteria decision making methodology (MCDM) and geographic information system (GIS) techniques have been widely applied to resilience assessment, empirical studies for regional transportation systems under complex climatic conditions are still relatively scarce, especially in Northeast China, a region with a variable climate and frequent extreme weather events, where there is a lack of targeted research frameworks and validation of methods.

In order to fill the research gaps mentioned above, this paper constructs a transportation system resilience assessment framework for multi-hazard scenarios, combines the MCDM and GIS technologies, and takes into full consideration of Changchun's climatic characteristics and transportation network features, and comprehensively evaluates the transportation system from the four dimensions of physical vulnerability, social exposure, disaster risk, and emergency response and recovery capability. By improving the hierarchical analysis method (AHP) and entropy weighting method, this paper effectively reduces the influence of subjectivity in the allocation of indicator weights, and reveals the resilience variability of the transportation system in different regions through spatial autocorrelation analysis. Based on the assessment results, this paper further proposes infrastructure improvement and emergency management optimization strategies for the low resilience region of Changchun, which provides scientific basis and practical guidance for improving the coping capacity of the transportation system in Northeast China and other similar climate regions.

Materials and methods

The study area

With the intensification of global climate change, the frequency and intensity of extreme weather events continue to increase, and the impact on the urban transportation system is becoming more and more serious, especially in areas with complex climate and frequent occurrence of extreme weather. Changchun, as an important city in Northeast China, is located in a temperate continental climate zone with long, cold winters and short, rainy summers, and this special climatic feature makes its urban transportation system susceptible to extreme weather events such as torrential rains, snowstorms, and high winds. According to the meteorological data statistics of Changchun over the past ten years, extreme weather events have shown a trend of increasing frequency (as shown in Table 1). For example, on August 28, 2012, a typhoon triggered 117.3 mm of heavy rainfall, leading to extensive flooding and nearly paralyzing urban transportation; a blizzard in December 2020 made the thickness of snow up to 30 cm, making several main roads impassable and limiting public transportation, which severely impacted residents' commuting. In addition, transportation hubs in low-lying areas of Changchun are particularly vulnerable to extreme rainfall events, often experiencing severe traffic congestion and vehicle damage due to waterlogged roads. These examples reflect the significant vulnerability of Changchun's transportation system to extreme weather, which is mainly characterized by road damage, sudden reduction in traffic flow, and long recovery time. In view of this, it is of great scientific significance and practical value to systematically assess the resilience capacity of the Changchun transportation system, reveal its vulnerability differences in different regions, and propose targeted enhancement strategies. The study area is shown in Fig. 1.

Indicator system and data sources

In order to comprehensively assess the resilience of the transport system in Changchun City, this study carefully selected a series of criteria and indicator layers based on four core dimensions: hazard, exposure, vulnerability and emergency response and recovery capability. The selection of these indicators aims to multi-dimensionally analyses the performance of the transport system under extreme weather, reveal its vulnerabilities and assess its coping and recovery capabilities.

Firstly, the selection of criteria layers is based on the internationally recognized Disaster Resilience Assessment Framework, which integrates the potential threat of natural hazards, the vulnerability of the system, the degree of exposure, and its emergency response and recovery capacity^{22,23}. Based on the characteristics of natural hazard

Times	Selected cases of extreme rainfall, snowfall and impacts
December 2011	In mid-December, successive snowfalls led to heavy snow accumulation and traffic congestion in the city, and some bus routes were suspended, making commuting difficult
August 2012	On 28 August, 117.3 mm of rain fell as a result of the typhoon, causing most of the roads to become waterlogged, paralyzing traffic and stranding a number of vehicles
June 2013	On 27 June, a short period of heavy rainfall led to flooding in low-lying areas of the city and backwash in drainage wells, seriously affecting traffic flow
January 2015	From 26 to 27 January, heavy snowfall caused disruptions at a number of transport hubs and massive delays at airports and railway stations, affecting passenger travel
February 2016	Between 25 and 26 November, snowfall covered the main roads, preventing many vehicles from starting because of the snow and virtually paralyzing traffic in the city
August 2018	On 14 August, localized waterlogging and flooding occurred, resulting in traffic congestion and impassability for pedestrians and vehicles
June 2019	On 2 June, the maximum rainfall reached 125.0 mm, and the entrance of Northeast Normal University was severely flooded, disrupting traffic and flooding many cars
December 2020	In late December, snowfall led to the accumulation of snow up to 30 cm deep in many sections of the city, making it difficult for vehicles to pass through and traffic control measures were implemented
July 2021	From 6 to 7 July, the rainfall resulted in 48 large waterlogged intersections and road sections across the city, severely affecting the traffic system and making access difficult

Table 1. Selected cases of extreme rainfall, snowfall and impacts.

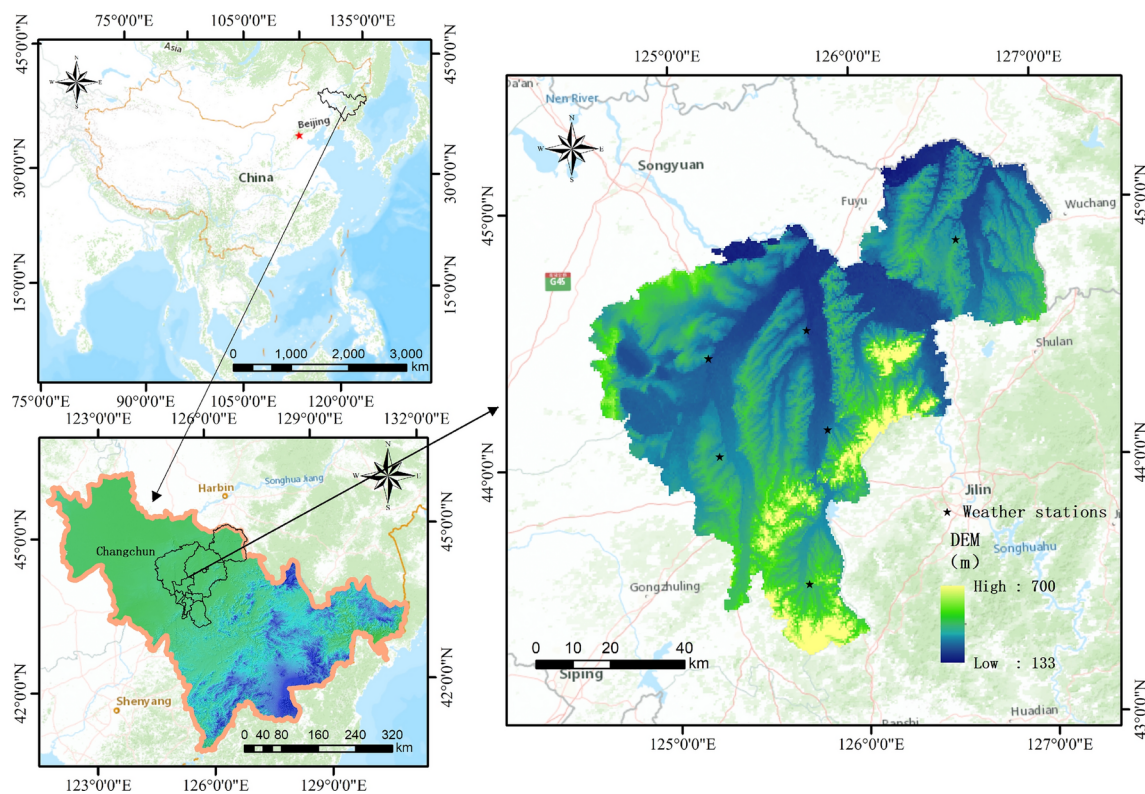


Fig. 1. Location map of the study area (This map was generated using ArcGIS 10.8 (ESRI <https://www.esri.com/en-us/home>)).

risks faced by Changchun (e.g., rainfall, snowstorms, strong winds, and extreme low temperatures), these criteria layers are effective in revealing key vulnerabilities of the transport system in the context of a disaster and provide targeted insights for policy makers²⁴.

On this basis, an indicator layer is used to accurately reflect each key aspect of the transport system in Changchun. Each indicator incorporates actual data and is consistent with widely used international assessment methods. Specifically, the Hazard Dimension: by analyzing historical data on disasters in Changchun, indicators such as ‘maximum daily rainfall’ and ‘snowfall intensity’ are selected to assess the potential threat of natural disasters to the transport system. Exposure Dimension: Indicators such as ‘Population Density’ and ‘Road Network Density’ are selected to reveal the degree of exposure of the transport system and the people it serves to disasters. Vulnerability Dimension: Measures the vulnerability of the system by reflecting the stress capacity of the transport network and the risk of damage during extreme events through ‘building density’ and ‘traffic flow’. Disaster prevention and mitigation capacity dimension: indicators such as ‘distribution of emergency facilities’

and ‘reserve of relief materials’ are used to assess the response capacity of the transport system in Changchun in the event of a disaster and the potential for post-disaster recovery. In addition, indicators such as the ‘GDP contribution rate’ measure the impact of the disaster on the overall economic vitality of the city and the speed of recovery.

These indicators are not only closely related to the actual situation in Changchun, but also provide a specific basis for improving the resilience of the urban transport system, ensuring scientific and targeted decision-making. Specific indicators and data sources are shown in Table 2 (the process of data calculation is shown in the supplementary material), and Fig. 2 shows the technical route and methodology used in this study. The maps were generated using ArcGIS 10.8 (ESRI, <https://www.esri.com/en-us/home>). All maps were created by the authors, using data from public datasets and original calculations.

Methodology

Improved AHP method

The improved AHP method is based on the original method and introduces a ‘triple scale’ to construct the comparison matrix. The triple scale simplifies the comparison process compared to the traditional 9-level scale, making it more intuitive and efficient for decision makers to determine the importance of each factor. This improvement addresses the difficulty of subjective judgement, which is common in traditional AHP, and improves the consistency and accuracy of judgement by reducing the complexity and cognitive load. Especially when dealing with the complex traffic resilience assessment of Changchun City, the triple scaling method can better adapt to the multi-criteria decision-making scenarios of practical applications, and ensures that the decision-making results are scientific and practical^{25,26}. The basic steps are as follows:

- 1. Construction of a three-level structural model. The three-level structural model consists of: the target level, the standard level and the indicator level.
- 2. Establish the comparison matrix A_{ij} . Each layer of indicators was compared two by two to create a comparison matrix A_{ij} .

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nm} \end{bmatrix} \tag{1}$$

Standard layer	Indicator layer	Data Type	Resolution/time	Data Source
Hazard	Average annual rainfall	Raster data	30 m/2017–2022	National Data Center for Meteorological Sciences
	Maximum daily rainfall	Raster data	30 m/2010–2020	China Meteorological Administration
	Accumulated snowfall	Raster data	30 m/2017–2022	National Data Center for Meteorological Sciences
	Daily minimum temperature	Raster data	30 m/2017–2022	National Data Center for Meteorological Sciences
	Maximum daily wind speed	Raster data	2017–2022	National Data Center for Meteorological Sciences
Vulnerability	Number of cars per 10,000 people	Attribute data	30 m/2020	Geospatial data cloud
	1-h traffic circle coverage	POI	30 m/2022	Planning cloud
	2-h traffic circle coverage	POI	30 m/2022	Planning cloud
	Road network density	Road network shape file	30 m/2022	Geospatial data clouds
	Building density	POI	30 m/2022	Planning cloud
	Reachability	Raster data	30 m/2022	Geospatial data clouds
	Percentage of transportation expenses	Attribute data	2022	Changchun Statistical Yearbook
	Educational attainment	Attribute data	30 m/2022	Changchun Statistical Yearbook
	Population density	Attribute data	100 m/2022	Geospatial data clouds
	Traffic hub density		30 m/2022	Planning cloud
Exposure	Vulnerable Populations	Attribute data	100 m/2022	Changchun Statistical Yearbook
	LULC	Raster data	30 m/2022	databox
	Average daily traffic	Attribute data	2022	Changchun Statistical Yearbook
Emergency response and recovery capability	GDP per capita	Attribute data	1 km/2022	National Bureau of Statistics
	Transportation GDP contribution	Attribute data	2022	Changchun Statistical Yearbook
	NDVI	Landsat 8 OLI/TIRS	30 m/2022	Databox
	Distribution of relief supplies	POI	30 m/2022	Planning cloud
	Number of doctors per 10,000 people	Attribute data	1 km/2022	Changchun Statistical Yearbook

Table 2. Selection of urban transportation system resilience indicators and data sources.

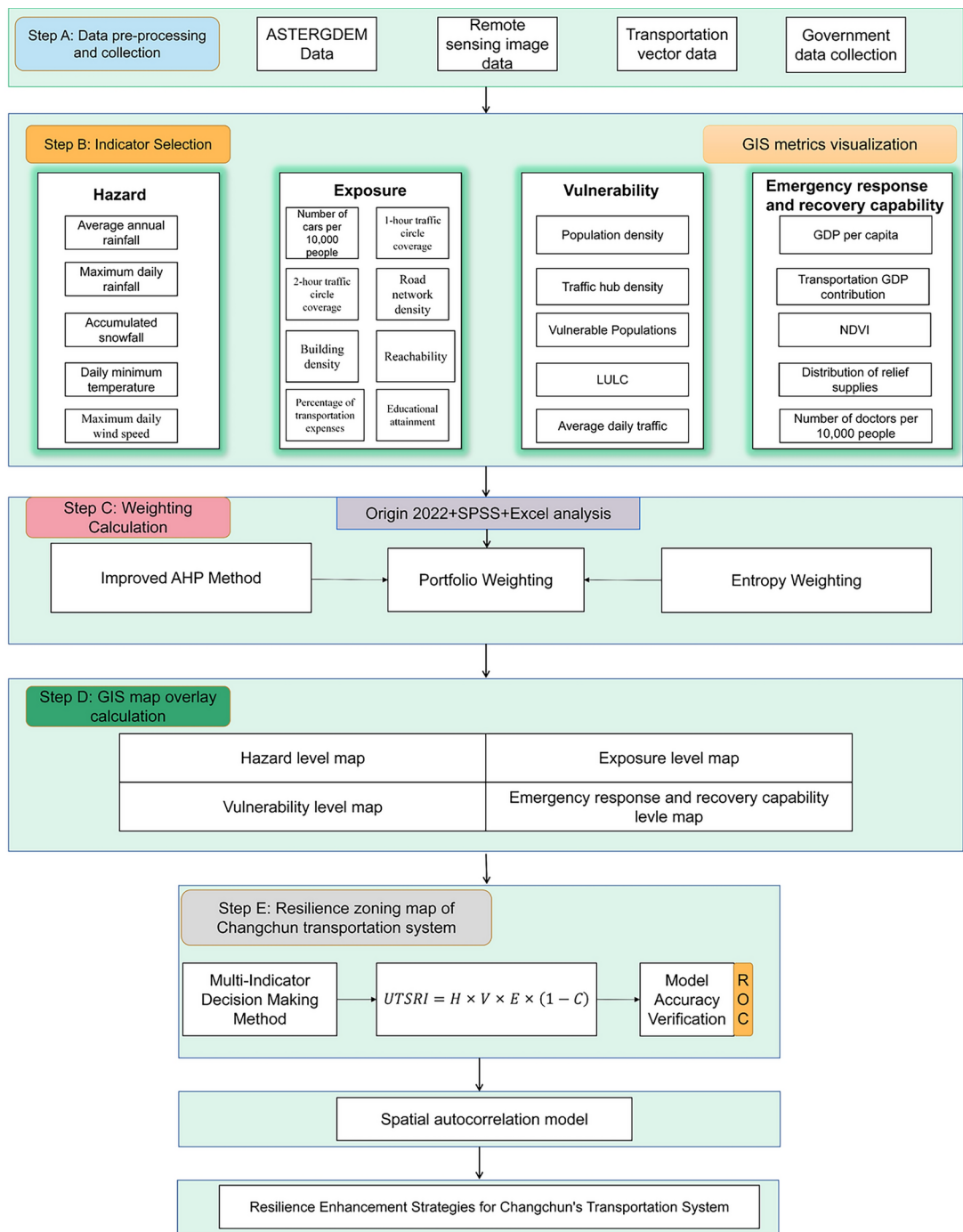


Fig. 2. Flow chart of urban transportation system resilience evaluation.

where A_{ij} is the importance obtained by the i -th element compared to the j -th element, the specific meaning is shown in Table 3.

3. Solve for the ranking index r_i . This is calculated by summing the results of the comparison of the i th criterion with the other criteria.

$$r_i = \sum_{j=1}^n A_{ij} \quad (2)$$

Retrieve a value	Hidden meaning
0	Indicator A_i is less important than A_j
1	Indicator A_i is equal in importance to A_j
2	Indicator A_i is more important than A_j

Table 3. Scale of significance.

4. Construct the judgment matrix B_{ij} . According to the sorting index r_i , use Eq. (3) to solve for the element b_{ij} , and construct the judgment matrix B_{ij} .

$$b_{ij} = \begin{cases} \frac{r_i - r_j}{r_{max} - r_{min}} k_m - 1 + 1 & r_i \geq r_j \\ \left[\frac{r_i - r_j}{r_{max} - r_{min}} k_m - 1 + 1 \right]^{-1} & r_i \leq r_j \end{cases} \quad (3)$$

Formula: $r_{max} = \max(r_i)$, $r_{min} = \min(r_i)$, $k_m = \frac{r_{max}}{r_{min}}$.

5. Solve for the transfer matrix c_{ij} . The transfer matrix c_{ij} is solved using the following equation

$$c_{ij} = \lg b_{ij}, (i, j = 1, 2, \dots, n) \quad (4)$$

6. Solve for the optimal transfer matrix D_{ij} . D_{ij} is solved using Eq. (5)

$$d_{ij} = \frac{1}{n} \sum_{k=1}^n (C_{ik} - C_{jk}) \quad (5)$$

7. Solve for the fitted consistency matrix B_{ij}^t . Solve using Eq. (6).

$$b_{ij}^t = 10^{d_{ij}} \quad (6)$$

8. Solve for the eigenvectors of B_{ij}^t , using the summation method.

9. Then the weights of the indicators obtained by the improved AHP are

$$\omega_j = a \cdot (\omega_1, \omega_2, \dots, \omega_n)^T \quad (7)$$

Entropy method

Entropy was first introduced into information theory by Shannon, and has been very widely used in various fields^{27,28}. The entropy weighting method solves the need for objectively determining weights in multi-indicator decision-making by calculating the information entropy of each indicator to determine its weight. In this study, the entropy weighting method is used to reduce the influence of human subjective judgement on the weights of indicators and to ensure objectivity based on data. Specifically, information entropy is used to measure the amount of information and uncertainty contained in the indicator, and the more informative the indicator (i.e., the higher the information entropy), the greater the weight in the decision. Compared with the traditional assignment method, the entropy weighting method can more accurately reflect the impact of each indicator on the assessment of traffic resilience, especially in the complex multi-hazard context.

The calculation steps of the entropy weighting method include: firstly, standardizing each indicator, then calculating the information entropy of each indicator, and finally determining the weight of each indicator according to the entropy value. This method effectively avoids the subjectivity of human-assigned weights and provides a scientific basis for the multi-indicator decision-making in this study.

1. Construct the normalized initial matrix.

Suppose that n independent metrics are used to evaluate m urban transportation system resilience

$$X = (x_{ij})_{n \times m} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}_{n \times m} \quad (8)$$

where x_{ij} is the primary data value for the j -th standardized stratum under the i -th indicator.

- To ensure comparability across indicators, the data were normalized using the following formula, which takes into account differences in magnitude, size, and range at the indicator level. This process involves converting values to a scale of 0 to 1, regardless of whether they vary positively or negatively. This allows us to effectively eliminate any differences in magnitude, size, or range between indicators.

Positive indicators

$$x'_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (9)$$

Negative indicators

$$x'_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} \quad (10)$$

where $\max(x_j)$ and $\min(x_j)$ denote the maximum and minimum values under i for the same indicator, respectively, x_{ij} denotes the value taken by the i -th unit in the j -th sample, and x'_{ij} denotes the j -th indicator for the i -th given unit after normalization.

- Calculate the weight of the j -th indicator of the i -th node p_{ij} :

$$P_{ij} = \frac{x'_{ij}}{\sum_{i=1}^m x'_{ij}} \quad (11)$$

where P_{ij} is the j -th indicator weight in the i -th standardized stratum, $0 \leq p_{ij} \leq 1$.

- Calculate the information entropy of the j -th evaluation indicator:

$$e_j = -\frac{1}{\ln m} * \sum_{j=1}^m p_{ij} \ln p_{ij} \quad j = 1, 2, \dots, m \quad (12)$$

When $p_{ij} = 0$ or $p_{ij} = 1$, $p_{ij} \ln p_{ij} = 0$.

- Calculate the coefficient of variation d_j and the weight ω_j of each node index

$$d_j = 1 - e_j \quad (13)$$

$$\omega_j = \frac{d_j}{\sum_j d_j} \quad (14)$$

where d_j is the coefficient of variation term for the j -th indicator, ω_j is the weight of the j th indicator.

Combined weighting determination

In order to make the degree of difference between the three-scaled hierarchical analysis method, entropy weight method and the weights obtained and their corresponding distribution coefficient difference consistent, this study uses the distance function set three-scaled hierarchical analysis method and the improvement of the distance function of the weight method as $d(\omega_j, \omega_j)$ the distribution coefficients of the weights of both for α and β , then

$$d(\omega_j, \omega_j)^2 = \sqrt{\sum_{j=1}^n (\omega_j - \omega_j)^2} \quad (15)$$

$$d(\omega_j, \omega_j)^2 = (\alpha - \beta)^2 \quad (16)$$

For the composite weight λ_j of evaluation indicator j , the value is a linear weighting of the weights of the two above, i.e.

$$\lambda_j = \alpha\omega_j + \beta\omega_j \tag{17}$$

The weights of (15)-(17) are combined and solved to obtain the composite weights, in this study, subjective and objective weights are equally important, $\alpha = \beta = 0.5$. The weights of each indicator layer are calculated as shown in Table 4.

Multi-criteria integrated evaluation method

In this paper, the urban transportation system resilience index (UTSRI) is constructed using the "H-E-V-C" assessment framework of the comprehensive risk assessment theory of natural hazards²⁹, which is used to measure the coping ability and recovery ability of the urban transportation system in the face of natural disasters. In addition, the urban transportation system hazard (H) is calculated according to the logistic regression model³⁰, and the formula for calculating the urban transportation system hazard is shown in Eq. (18):

$$H = \frac{\exp(\beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_mx_m)}{1 + \exp(\beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_mx_m)} \tag{18}$$

where H is the urban transportation system hazard, $H \in [0, 1]$, the smaller the value of H, the lower the urban transportation system hazard; x_m is the independent variable and each indicator; β_m is the logistic regression coefficient.

The standard tier formulas for Exposure (E), Hazard (V), and Emergency response and recovery capability (C) are as follows.

$$E = \sum_{i=1}^n W_{ei} X_{ei} \tag{19}$$

$$V = \sum_{i=1}^n W_{vi} X_{vi} \tag{20}$$

$$C = \sum_{i=1}^n W_{ci} X_{ci} \tag{21}$$

where: The values of E, V, C and c correspond to exposure, vulnerability, and Emergency response and recovery capability, respectively; n is the total number of indicators; i is the i-th indicator, W_{ei} , W_{vi} , W_{ci} are weights of

Target Layer	Standard layer	Standard tier weights	Indicator layer	Indicator layer weights
Urban transportation system resilience	Hazard	0.238	Average annual rainfall	0.050
			Maximum daily rainfall	0.053
			Accumulated snowfall	0.047
			Daily minimum temperature	0.045
			Maximum daily wind speed	0.048
	Vulnerability	0.327	Number of cars per 10,000 people	0.042
			1-h traffic circle coverage	0.027
			2-h traffic circle coverage	0.027
			Road network density	0.054
			Building density	0.048
			Reachability	0.059
			Percentage of transportation expenses	0.031
			Educational attainment	0.034
	Exposure	0.219	Population density	0.056
			Traffic hub density	0.050
			Vulnerable Populations	0.042
			LULC	0.033
			Average daily traffic	0.038
	Emergency response and recovery capability	0.216	GDP per capita	0.051
			Transportation GDP contribution	0.043
			NDVI	0.037
			Distribution of relief supplies	0.044
			Number of doctors per 10,000 people	0.041

Table 4. Urban transportation system resilience index weights.

each indicator obtained by combined weighting determination respectively; X_{ei} , X_{vi} , X_{ci} are the quantitative values of indicators corresponding to exposure, vulnerability, and disaster prevention and mitigation capacity.

Modeling of urban transportation system resilience index

In this study, urban transportation system resilience index (USTRI) was positively correlated with transportation system hazard (H), exposure (E) and vulnerability (V), and negatively correlated with emergency response and recovery capability (C).

USTRI is calculated as follows:

$$USTRI = H \times E \times V \times (1 - C) \quad (22)$$

Spatial autocorrelation model

On the basis of the above calculation of the resilience level of the urban transport system, this paper analyses the spatial autocorrelation at different spatial scales using global Moran's I and local Moran's I. Global Moran's I is used to measure the overall spatial correlation of traffic flow in Changchun, revealing the aggregation or discrete nature of traffic flow distribution in the whole city scale; while local Moran's I is used to analyses the spatial correlation of traffic flow in local areas, identifying whether there is significant spatial aggregation of traffic flow in a specific local area in Changchun³¹. These two spatial autocorrelation indicators can comprehensively reveal the spatial distribution characteristics of traffic flow in Changchun and its variability in different regions.

The global Moran's I index uses the following equation:

$$I = \frac{n}{S_0} \frac{\sum_{i=1}^n \sum_{j=1}^n W_{ij} (x_i - u)(x_j - u)}{\sum_{i=1}^n (x_i - u)^2} \quad (23)$$

The local Moran's I index uses the following equation:

$$I_i = \frac{(x_i - u) \sum_{j=1}^n W_{ij} (x_j - u)}{\frac{1}{n} \sum_{i=1}^n (x_i - u)^2} \quad (24)$$

where x_i and x_j are the number of traffic nodes at the i -th and j -th traffic nodes, respectively; n , u , W_{ij} and S_0 are the traffic flow, the average value of traffic flow, the spatial weight matrix and the sum of all the weights, respectively. For global Moran's I, if $I > 0$, there is a strong positive correlation of traffic flow in space, if $I < 0$, there is a strong negative correlation of traffic flow in space, if $I = 0$, it means that traffic flow is basically randomly distributed in space. Local Moran's I Similar to global Moran's I.

Evaluation system

This chapter aims to analyses in detail the resilience of urban transport systems in Changchun in the face of natural disasters through a multidimensional assessment framework. GIS and MCDM, this study reveals the performance of the transport system and its weaknesses under extreme weather by comprehensively assessing the four core dimensions of the transport system: hazards, vulnerability, exposure, and emergency response and recovery capability. Against the backdrop of rapid urbanization and the increasing frequency of extreme weather events, urban transport systems are under immense pressure. Transport infrastructures not only need to meet daily operational needs, but also need to be able to cope with natural disasters³². To ensure the continuity and safety of the transport system, it is crucial to systematically assess its adaptive and resilience capabilities in disasters.

In order to comprehensively assess the resilience of the transport system in Changchun City, this study starts from four dimensions: first, the disaster dimension analyses the natural threats to the transport system, including the damage that may be caused to roads and transport facilities by extreme rainfall, snowfall, and strong winds; second, the vulnerability dimension evaluates the resilience of the transport infrastructure in the face of a disaster, with an emphasis on the analysis of the key elements such as the traffic flow, the density of the road network, and the density of buildings; third, the exposure dimension evaluates the resilience of transport infrastructure to disasters, focusing on the analysis of traffic flow, road network density, building density and other key elements; again, the exposure dimension focuses on population density, location of transport hubs, and distribution of disaster-prone people to reveal which areas or facilities are exposed to higher risks under natural disasters; finally, the emergency response and recovery capability dimension examines Changchun's ability to manage emergencies in the face of a disaster and the speed of post-disaster recovery, including distribution of materials, medical resources, and economic capabilities.

By evaluating these four dimensions, this chapter not only reveals the weaknesses of Changchun's transport system, but also provides a basis for improving the system's resilience. The goal of this study is to provide scientific decision support for urban planners and policy makers to help develop more adaptive and disaster-resilient transport planning strategies, thereby enhancing the ability of Changchun's transport system to cope with the challenges of extreme weather and uncertainty in the future.

Urban transportation system hazard evaluation

The hazard dimension evaluates the external natural threats that the transportation system faces. Changchun's urban transport is vulnerable to extreme weather conditions, particularly heavy rainfall, snowstorms, and strong winds. By assessing these hazards, this study aims to quantify the potential impact of extreme weather on the city's transportation infrastructure and its operations. The selected indicators reflect key meteorological phenomena that pose the greatest risks.

Selection Rationale:

1. **Average Annual Rainfall:** The average annual rainfall in Changchun reflects the frequency and total amount of rainfall, which is directly related to the risk of flooding and the carrying capacity of the road drainage system. This indicator is used to assess the long-term impact of rainfall on low-lying areas and poorly drained roads³³.
2. **Maximum Daily Rainfall:** Extreme rainfall events are one of the main causes of transportation system paralysis³⁴, for example, the heavy rainfall in August 2012 led to the flooding and impassability of several transportation hubs in Changchun. This indicator selection helps to quantify the impact of extreme rainfall on road flooding and transportation disruption.
3. **Accumulated Snowfall:** Winter snowfall is one of the main challenges for the Changchun transportation system. Snow thickness directly affects road capacity and vehicle safety³⁵, as in the case of the 2020 blizzard event, which caused snow to accumulate up to 30 cm deep on city roads, severely hampering traffic operations.
4. **Daily Minimum Temperature:** Cold temperatures lead to icy road surfaces and increase the risk of traffic accidents. This indicator reveals the potential threat of cold weather to road conditions and capacity by quantifying the frequency and extent of low temperatures³⁶.
5. **Maximum Daily Wind Speed:** Strong winds may cause damage to infrastructure such as bridges and traffic signs, and affect traffic flow³⁷. For example, in Changchun City, the efficiency of public transit and non-motorized vehicle operations is usually significantly disrupted during strong winds. The spatial distribution of hazard indicators is shown in Fig. 3.

Urban transportation system vulnerability evaluation

The vulnerability dimension measures the internal weaknesses of the transportation system that make it more susceptible to failure during natural disasters. This includes assessing the robustness of the road network, traffic density, and the distribution of critical infrastructure. By identifying areas with high vulnerability, this dimension helps determine where improvements in infrastructure and urban planning are necessary to reduce the system's susceptibility to disruption.

Selection Rationale:

1. **Number of Cars per 10,000 People:** High vehicle density increases the likelihood of congestion, especially during emergencies. This indicator reflects the strain on transportation systems during high-traffic situations³⁸.
2. **(2) 1-h and 2-h Traffic Circle Coverage:** These indicators assess the reachability of essential services and transportation routes within certain timeframes, crucial for evacuation and emergency responses during disasters³⁹.
3. **Road Network Density:** A higher road network density can improve redundancy, allowing for alternative routes during blockages. It also highlights areas where the network might be too sparse to handle disruptions⁴⁰.
4. **Building Density:** Dense urban areas are more susceptible to disruptions because of the concentration of people and infrastructure, which may impede recovery efforts⁴¹.
5. **Reachability:** This assesses how easily people and goods can move across the city, which is vital for disaster response and recovery⁴².
6. **Educational Attainment:** This indicator indirectly reflects the population's capacity to respond effectively to disasters, as better-educated communities may be more prepared and informed about emergency protocols⁴³. The spatial distribution of vulnerability indicators is shown in Fig. 4.

Exposure evaluation of urban transportation system

The exposure dimension evaluates the extent to which people, infrastructure, and economic assets are exposed to natural hazards. High exposure means that a larger proportion of the population and critical infrastructure is at risk during extreme weather events. This analysis helps pinpoint areas that require more robust protection and disaster mitigation strategies.

Selection Rationale:

1. **Population Density:** Transportation needs are higher in high population density areas, and disruptions due to extreme weather may cause large-scale evacuation problems⁴⁴. High population density areas in the center of Changchun (e.g., Nanguan and Chaoyang districts) are important scenarios for the application of this indicator.
2. **Vulnerable Populations:** Elderly people, children and people with disabilities are more at risk in extreme weather. For example, some of the older communities in Changchun's urban and rural areas have a high concentration of vulnerable groups and weak transportation services and emergency response, making this indicator particularly critical for revealing social risks⁴⁵.

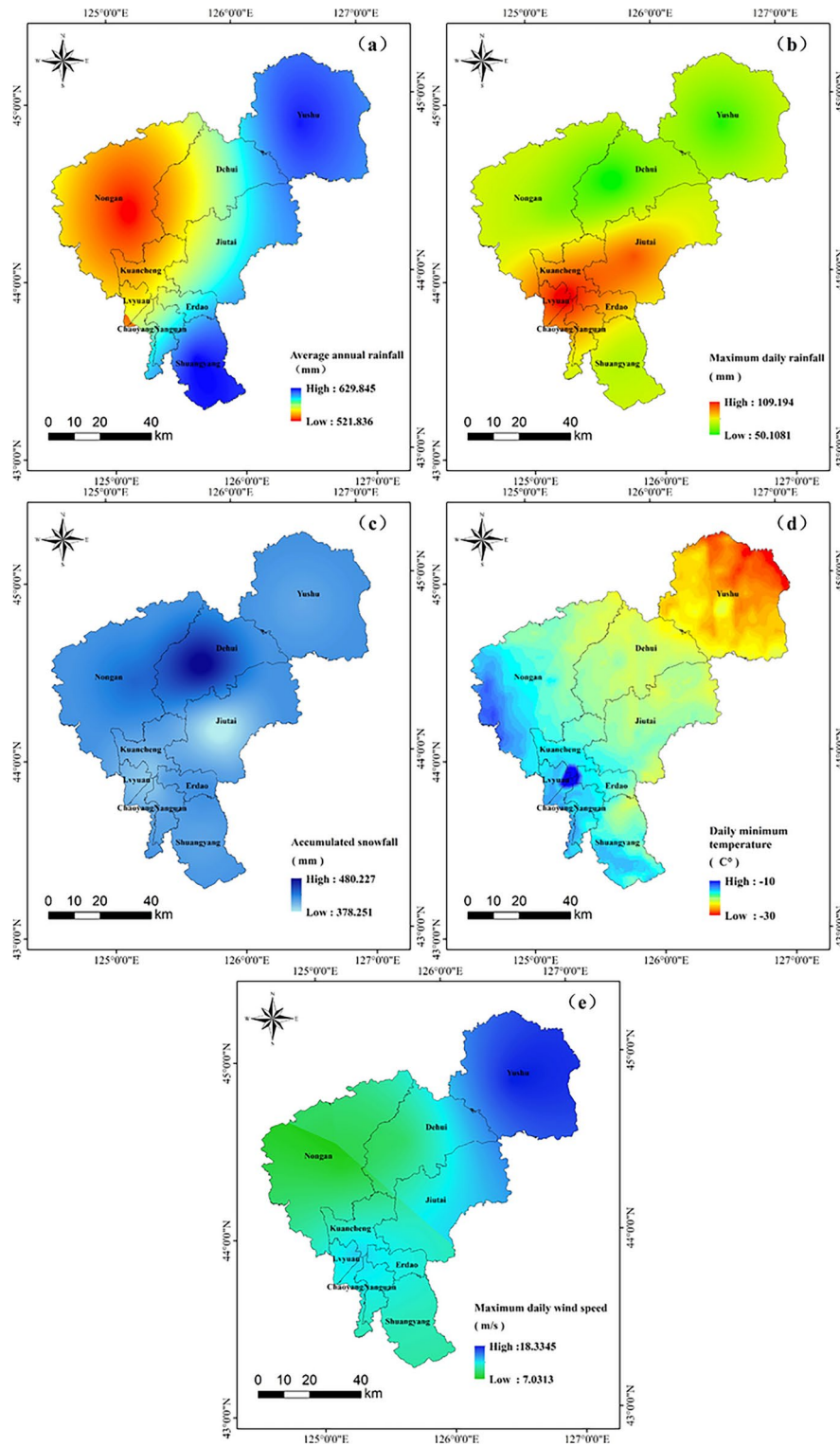


Fig. 3. Spatial distribution of hazards. (a) Average annual rainfall, (b) Maximum daily rainfall (c) Accumulated snowfall, (d) Daily minimum temperature, (e) Maximum daily wind speed.

3. Traffic Hub Density: Transportation hubs are key nodes of urban transportation networks, and their density directly affects the degree of impact of disasters on the operation of the entire network⁴⁶. Changchun Railway Station and major bus transfer hubs have shown to be highly exposed areas during historical extreme weather.
4. LULC (Land Use/Land Cover): Different forms of land use (e.g., urban green space, low-lying areas) directly affect the regional resilience to flooding and heavy rainfall⁴⁷. The trend of decreasing green space and urban

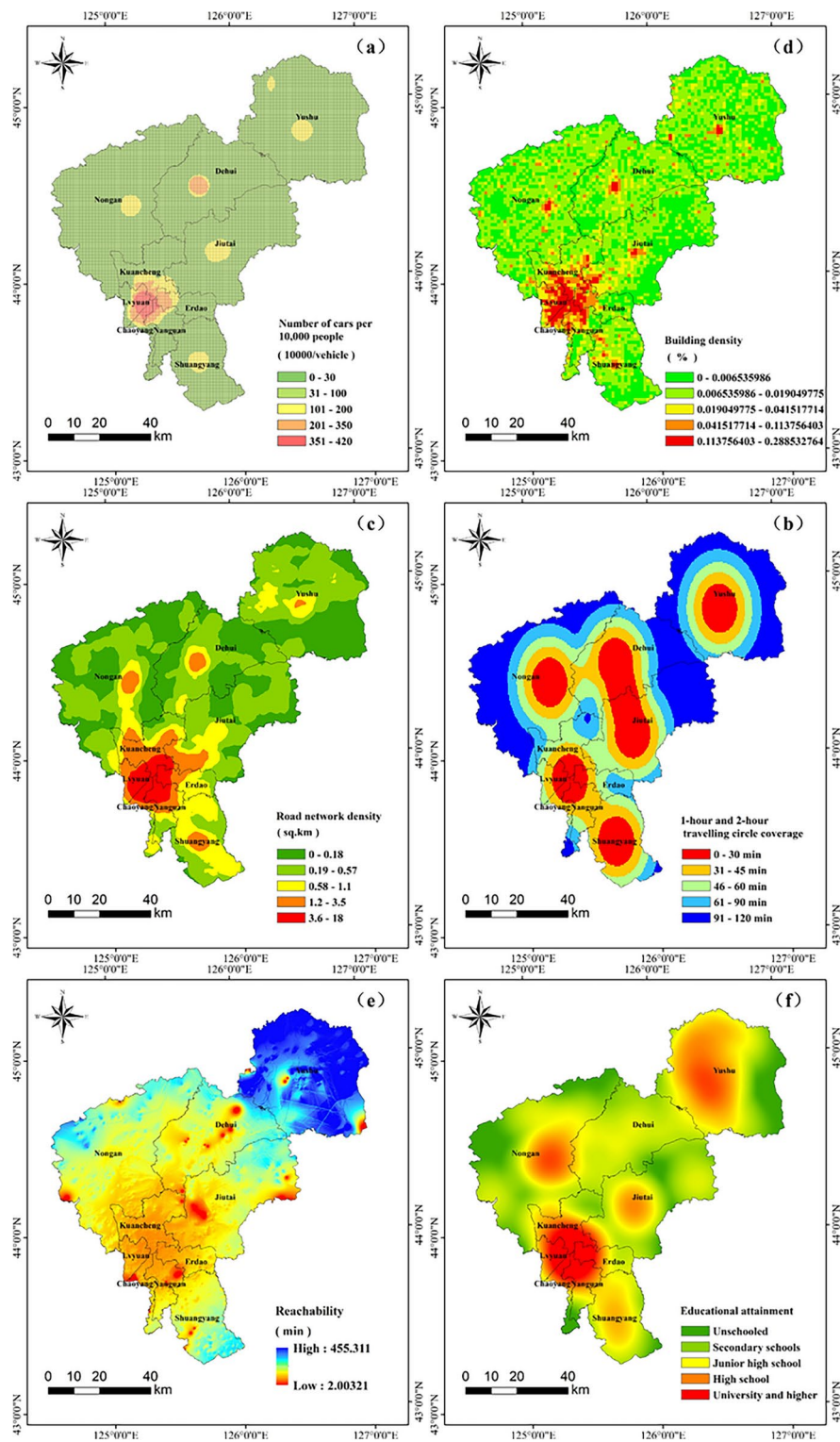


Fig. 4. Spatial distribution of vulnerability indicators. (a) Number of cars per 10,000 people, (b) 1-h and 2-h traffic circle coverage, (c) Road network density, (d) Building density, (e) Reachability, (f) Educational attainment.

expansion in Changchun city reinforces the importance of this indicator to assess the role of urban expansion in exacerbating exposure risk. The spatial distribution of exposure indicators is shown in Fig. 5.

Evaluation of emergency response and recovery capability of urban transportation system

The emergency response and recovery capability dimension assess how well the transportation system can manage and recover from disruptions caused by natural disasters. This dimension is critical for ensuring that the system can quickly bounce back from damage, minimizing downtime and maintaining essential services. The selected indicators reflect the system's preparedness and its ability to restore functionality.

Selection Rationale:

1. GDP per Capita: This indicator reflects the region's economic strength and its ability to fund disaster preparedness and recovery initiatives. A higher GDP per capita generally correlates with better infrastructure and emergency services, enhancing recovery capabilities⁴⁸.

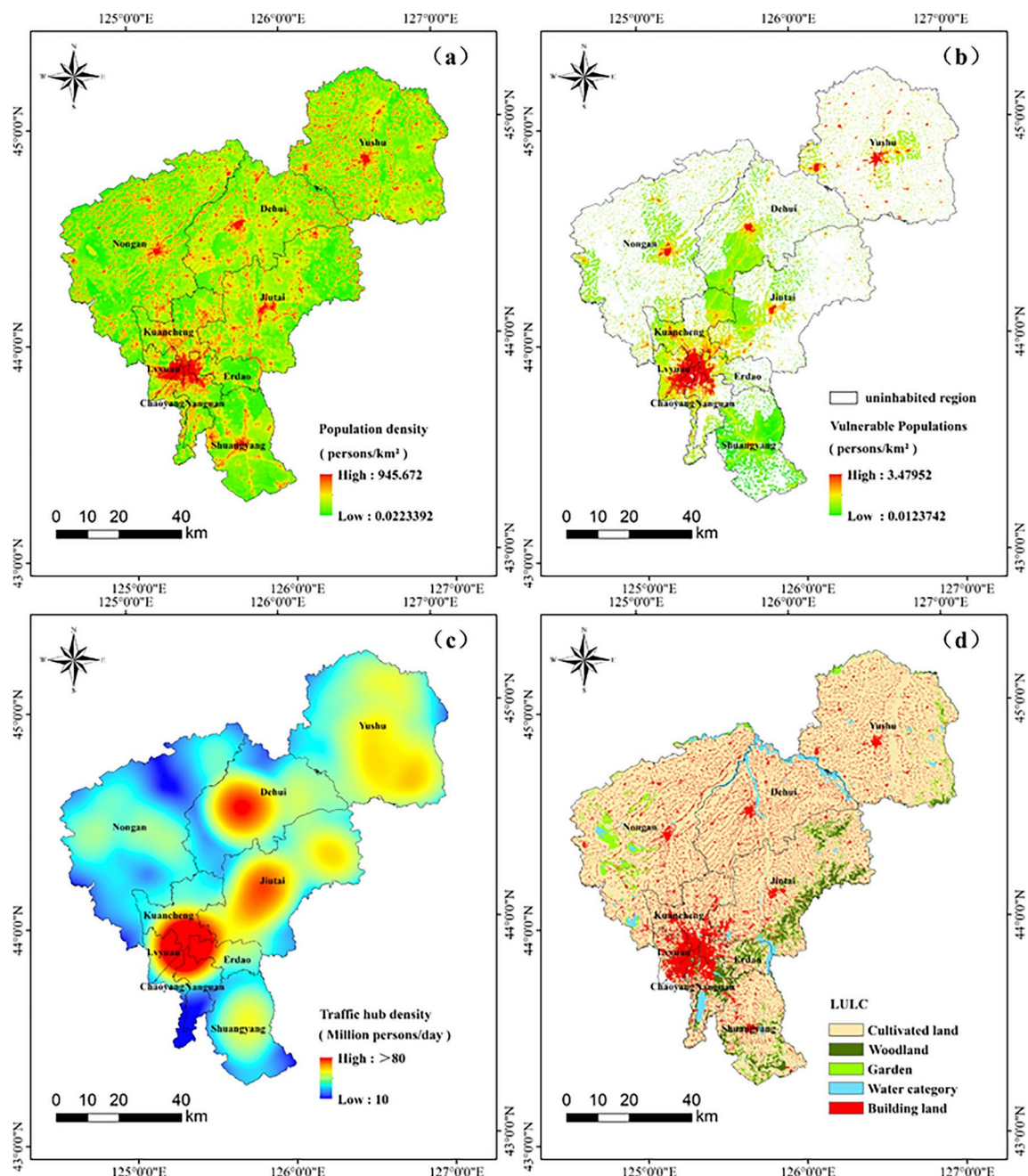


Fig. 5. Spatial distribution of expositional indicators. (a) Population density, (b) Vulnerable Populations, (c) Traffic hub density, (d) LULC.

2. NDVI (Normalized Difference Vegetation Index): Vegetation cover plays a role in mitigating flooding and erosion risks, helping to absorb excess water and stabilize the ground. Areas with higher NDVI values may be better equipped to handle natural disasters⁴⁹.
3. Distribution of Relief Supplies: The strategic placement of relief materials ensures that aid can be delivered promptly during emergencies, which is critical for reducing the impact of disasters on both the population and the transportation system⁵⁰.
4. Number of Doctors per 10,000 People: The availability of medical resources is essential for providing immediate care during and after disasters. Areas with more doctors can respond more effectively to health emergencies that arise during extreme weather⁵¹. The spatial distribution of emergency response and recovery capability indicators is shown in Fig. 6.

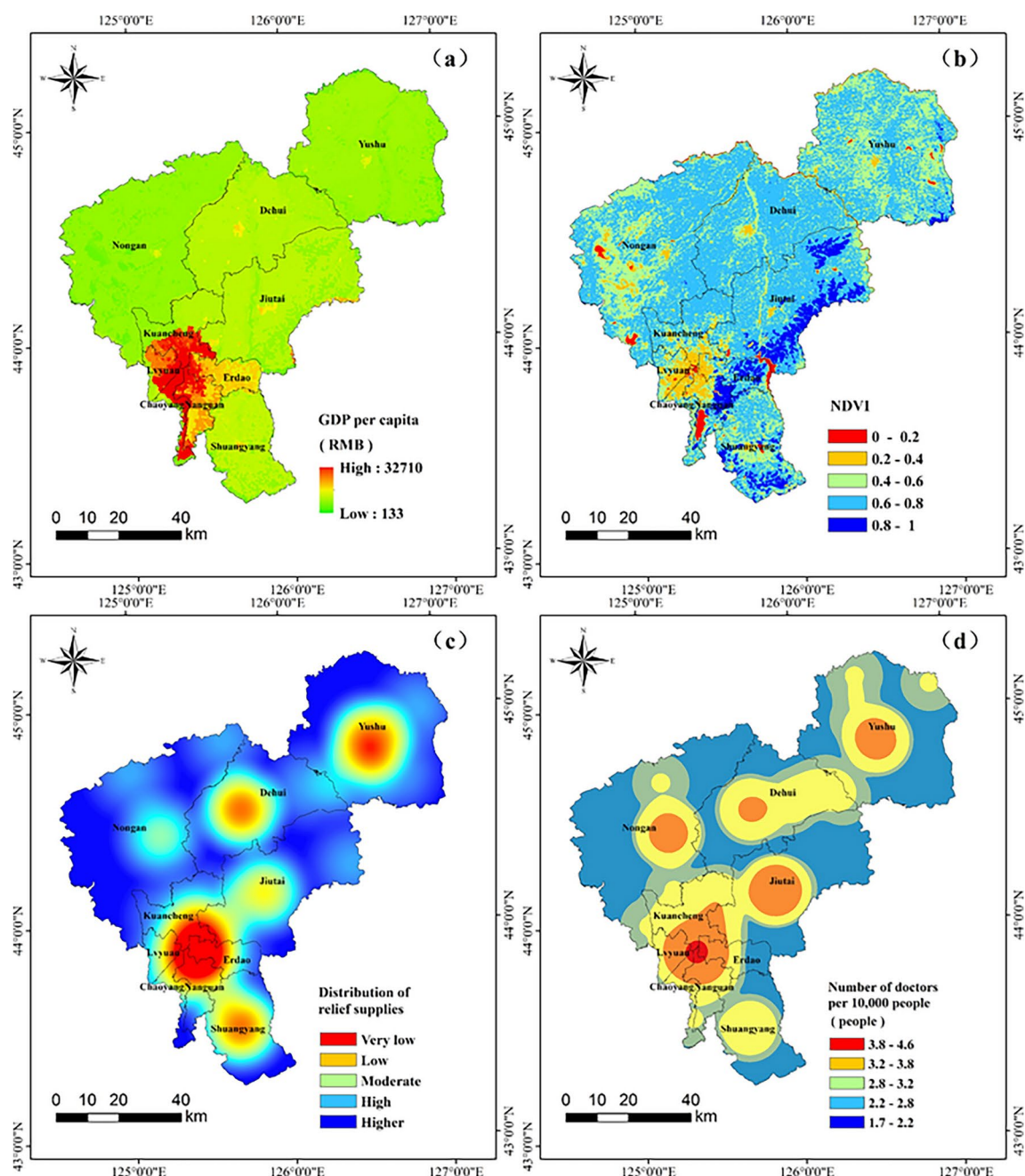


Fig. 6. Spatial distribution of emergency response and recovery capability indicators. (a) GDP per capita, (b) NDVI, (c) Distribution of relief materials, (d) Number of doctors per 10,000 people.

Transport system resilience assessment and its enhancement strategies

Evaluation system zoning map

Based on the analysis of the four dimensions, the transport system in Changchun City shows obvious regional differences when facing natural disasters. In the disaster dimension, the eastern region has a higher disaster risk, especially during heavy rain and snowstorms, which is mainly related to the lower terrain and developed water system, which is prone to flooding and traffic disruption. In contrast, remote areas are relatively low risk due to their sparse population. The vulnerability dimension reveals higher vulnerability of the transport system in city center areas such as Chaoyang and Nanguan districts, where high building densities and traffic flows are prone to traffic congestion and limited emergency response during extreme weather. Peripheral areas have lower vulnerability due to their lower population and building densities, lighter traffic loads, and more time and space for response. The exposure dimension shows that city center areas have the highest exposure, especially in transport hubs and densely populated areas, and areas with high exposure are at greater risk under natural hazards. The concentration of transport hubs and important public facilities makes these areas susceptible to severe impacts during extreme weather, and effective evacuation and emergency management measures are urgently needed. Analysis of the emergency response and resilience dimensions shows that urban centers are more resilient due to higher economic resources and better stockpiles of emergency supplies, while remote areas are less resilient, mainly due to their insufficient economic resources and medical relief systems. This difference highlights the imbalance in emergency response capacity between regions, suggesting that remote areas need more resources to improve disaster recovery capacity.

Taken together, the four-dimensional assessment reveals the weaknesses and key risk areas of Changchun's transport system in natural disasters, providing a strong basis for improving transport resilience. The specific partition map is shown in Fig. 7.

Resilient zoning and validation of the Changchun transport system

Using the UTSRI formula (22), a superimposed analysis of the four standard layers (Fig. 8) shows that the transport system across the region exhibits significant differences in resilience in the face of natural disasters. The very low resilience region (38.6%) is mainly located in the peripheral and rural areas of Changchun City, where the transport infrastructure is relatively poor and the network is sparse, with weak resilience and recovery capacity in the face of extreme weather, and slower recovery from transport disruptions. This may be related to the lack of emergency response resources in these areas as well as a single transport network. Low resilience areas (30.7 per cent) are mainly located in urban and rural areas, and although transport facilities are relatively good, these areas are still more vulnerable in the event of disasters and have a relatively low recovery efficiency due to the lack of strong emergency response mechanisms. Medium resilience regions (25.3%) are concentrated in urban areas and surrounding regions, which have more developed transport facilities and a certain degree of emergency response and recovery capacity, but still face certain challenges in extreme climates due to higher population density and high transport demand. Finally, high resilience areas (5.4%) are concentrated in city centers and major transport hubs, with well-developed transport systems, sufficient economic resources, and a high level of emergency response and rapid recovery capacity; these areas play a central role in urban transport resilience.

The higher resilience of urban centre areas is due to the fact that urban centres have rich economic resources and well-developed infrastructures, which make their transport systems more resilient in terms of emergency response and rapid recovery in the face of natural disasters. In addition, the clustering of transport hubs and the application of intelligent traffic management systems also enhance the resilience of transport networks. For example, during major rainfall or snowstorms, trunk roads and transport hubs in central areas are able to rely on redundant road networks to divert traffic flows quickly. In contrast, urban-rural and remote areas have insufficient redundancy and emergency response capacity in their transport systems due to relatively poor economic and infrastructure development. There are fewer transport hubs in these regions and the density of the transport network is lower, making the transport system slower to respond and recover in the event of a disaster. In addition, the uneven distribution of emergency response resources, such as material stockpiles and rescue forces, exacerbates the vulnerability of these regions.

In order to verify the reliability of the model results, this study used 63 traffic accident-prone and frequent points identified by the Changchun Traffic Safety Committee in 2022 and 151 congestion points based on Changchun's traffic GPS data and road blockage records over the past five years for validation. Figure 9 displays the map of the validation points. The ROC curve is employed to assess the performance of a dichotomous model, with each point representing the true rate and false positive rate under various classification thresholds⁵². Theoretically, this metric ranges from 0.5 to 1.0, where a larger area under the ROC curve indicates better model performance. Figure 10 illustrates the ROC curve for the subject's job characteristics. In the aforementioned validation, the area under the ROC curve is 0.89317, confirming the reliability and utility of the model.

Spatial autocorrelation analysis

On the basis of the above calculation of the resilience level of the urban transport system, this paper analyses the spatial autocorrelation at different spatial scales using the formulas of global Moran's I and local Moran's I in Section "Spatial autocorrelation model". Global Moran's I is used to measure the overall spatial correlation of traffic flow in Changchun, revealing the aggregation or discrete nature of traffic flow distribution within the city; local Moran's I is used to analyses the spatial correlation of traffic flow in local areas, identifying whether the traffic flow exhibits significant spatial aggregation or discrete phenomena in a specific area.

Based on the traffic flow data from 2012 to 2021, the value of global Moran's I increase from 0.42 to 0.51 (Fig. 11), indicating that the spatial autocorrelation of the urban transport system in Changchun City gradually increases. This change reflects that the spatial aggregation of traffic flow has been strengthened in that time

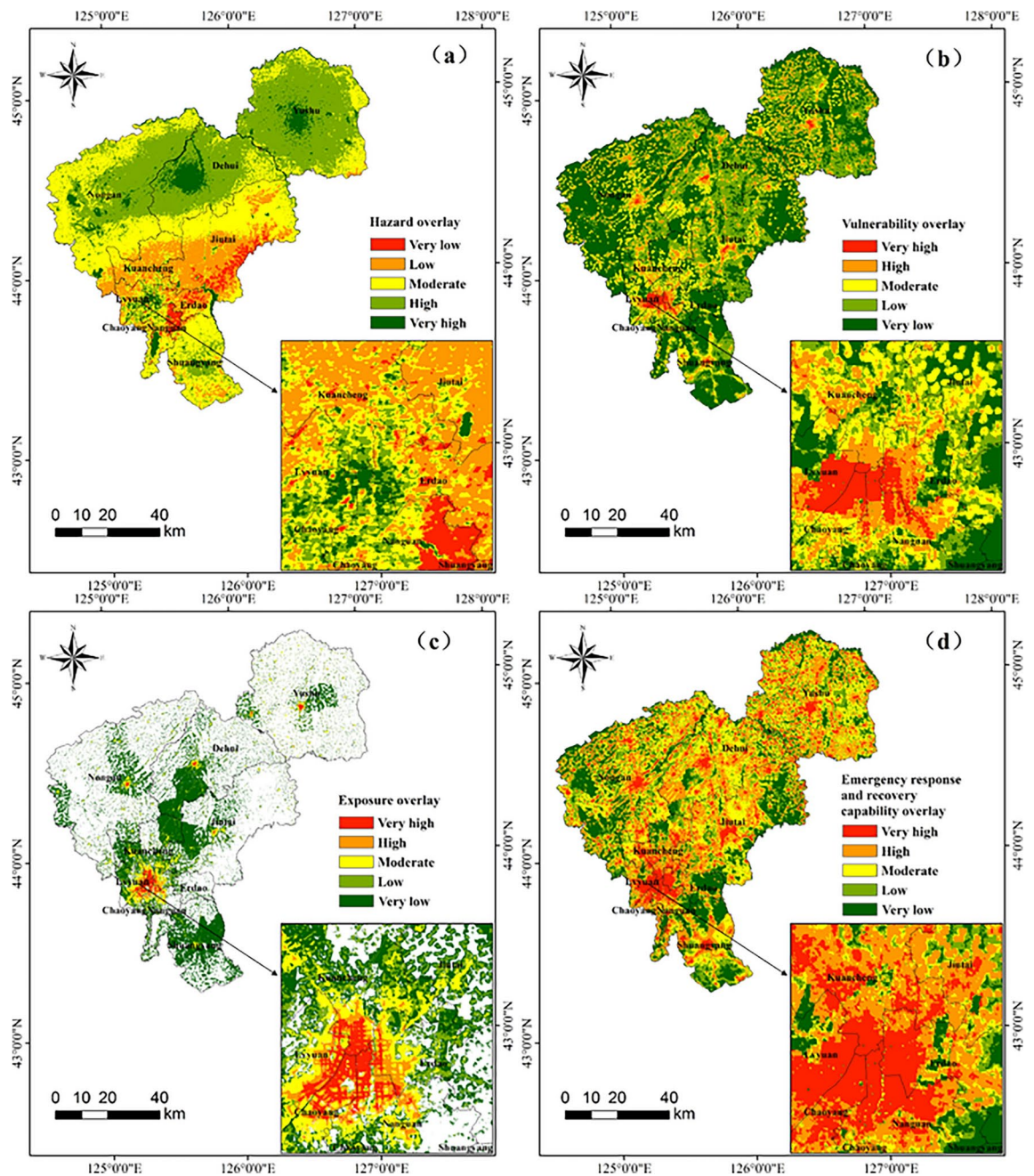


Fig. 7. Evaluation system zoning map. (a) Hazard zoning map, (b) Vulnerability zoning map, (c) Exposure zoning map, (d) Emergency response and recovery capability zoning map.

period, i.e., the concentration of traffic flow in a specific area has increased. It can be speculated that this trend is related to the transport infrastructure improvement measures implemented in Changchun in recent years, including the expansion of the road network and the optimization of the traffic management system. These measures have enhanced the adaptability of the transport system, enabling it to maintain relatively stable operation under changing traffic conditions.

Through local Moran's *I* analysis, combined with GIS clustering and outlier analysis tools, we further demonstrate the local spatial aggregation of traffic flows in Changchun. The High-high area represents the existence of high traffic pressure in this area and its adjacent areas during the disaster period, and is mainly located in the busy commercial centers of Chaoyang District, Nanguan District, Kuangcheng District and Luyuan District, which have a large number of commercial, shopping and recreational facilities and attract a large number of pedestrian and vehicular flows. At the same time, these areas are also densely populated residential areas with high traffic demand, but the carrying capacity of the surrounding road network is limited, leading to increased localized traffic congestion.

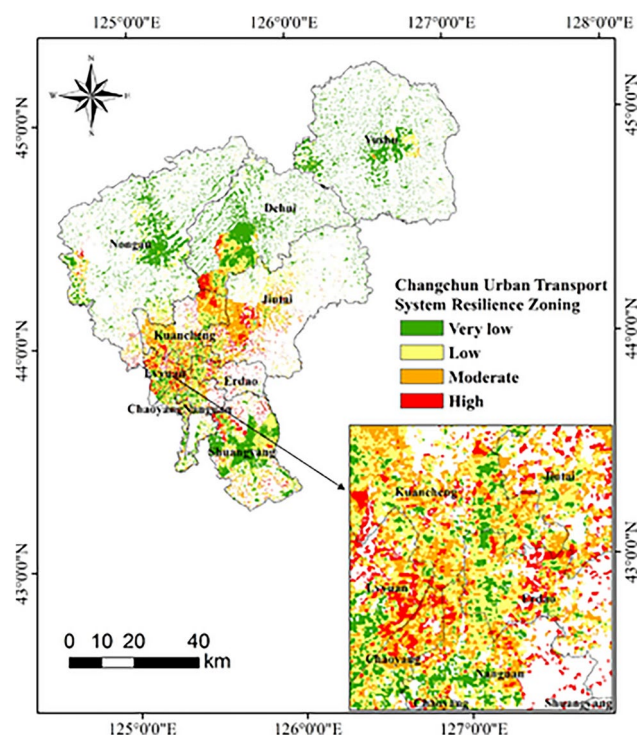


Fig. 8. Resilience zoning map of Changchun transportation system.

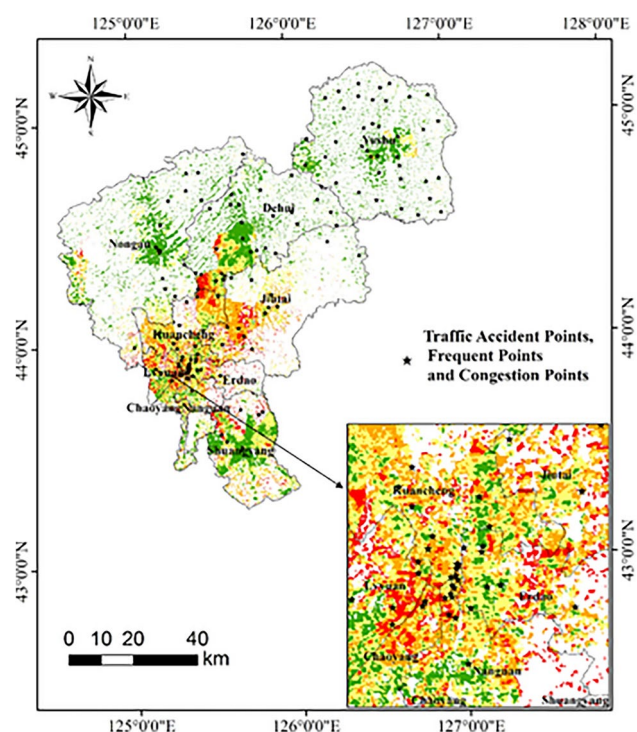


Fig. 9. Traffic accident-prone, multi-points map.

The Low-low areas, on the other hand, are concentrated in Yushu and Dewei in Changchun, which are farther away from the city center, with lower population density and relatively low traffic flow, presenting the characteristics of suburbs or urban-rural areas. Due to the sparse road network in these areas, traffic flows are less likely to gather locally and traffic pressure is low. The poor transport connectivity of these areas and the lack

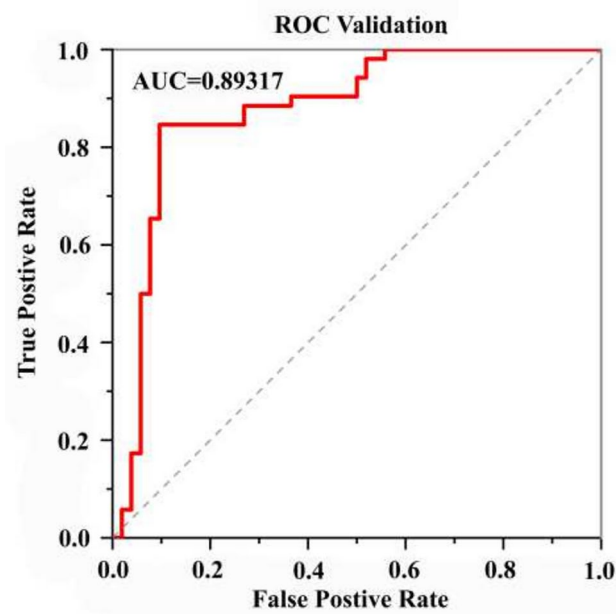


Fig. 10. Subject work characteristic (ROC) curve.

of convenient transport hubs and expressways further reduce the phenomenon of localized aggregation of traffic flows.

Resilience enhancement strategies

In order to enhance the overall resilience of the transport system in Changchun, a comprehensive multi-level and differentiated strategy should be adopted. First, for peripheral and rural areas (very low resilience regions), infrastructure construction needs to be strengthened, especially the upgrading of key facilities such as road reinforcement and drainage optimization, and the establishment of communication and emergency warning systems needs to be promoted in order to improve the resilience and post-disaster recovery efficiency of these regions. Second, optimizing emergency management and resource allocation is particularly important in the urban–rural interface (low resilience areas). By strengthening the emergency management system, establishing alternate routes and traffic flow dispersal mechanisms, as well as enhancing community participation and emergency education, the coping capacity of the transport system in these areas can be effectively enhanced under extreme weather conditions. For medium resilience areas in and around urban areas, the focus should be on promoting the application of intelligent traffic management systems, enhancing the resilience of infrastructure, and introducing green infrastructure development to mitigate the risk of urban flooding. In addition, in high resilience areas (e.g. city centers and transport hubs), existing strengths need to be maintained, resource concentration and emergency dispatch capabilities further strengthened, and technological innovations should be used to ensure the efficient resilience of the transport system in the event of a disaster. Finally, synergistic cooperation between regions should be promoted, regional linkage mechanisms should be established to ensure effective resource deployment, and post-disaster recovery mechanisms should be improved, especially in peripheral and urban–rural areas, in order to accelerate post-disaster rehabilitation of infrastructure and reduce long-term impacts. Through these comprehensive measures, Changchun's transport system will be more resilient and resilient in responding to extreme weather events, guaranteeing the continued operation and safety of urban transport.

Discussion

This study comprehensively assesses the performance of the Changchun transport system under natural disasters by constructing a multi-dimensional resilience assessment framework, which reveals the resilience differences between different regions. The results not only reflect the spatial heterogeneity of the transport system in Changchun under extreme weather conditions, but also highlight the significant differences in infrastructure and emergency management capabilities between regions.

The results of the assessment show that transport system resilience is higher in urban centers and relatively lower in urban–rural and remote areas. The underlying reason for this difference is the uneven distribution of economic resources and infrastructure. Urban centers are usually rich in resources and have well-developed transport infrastructure, and therefore have a stronger capacity to cope with and recover from disasters. On the other hand, urban–rural and remote areas have difficulty in responding quickly to the damage caused by disasters due to limited financial inputs and scarce transport hubs. In addition, urban centers are equipped with advanced emergency management systems that are able to channel traffic and dispatch resources through intelligent traffic management, thus enabling a quicker return to normal operations.

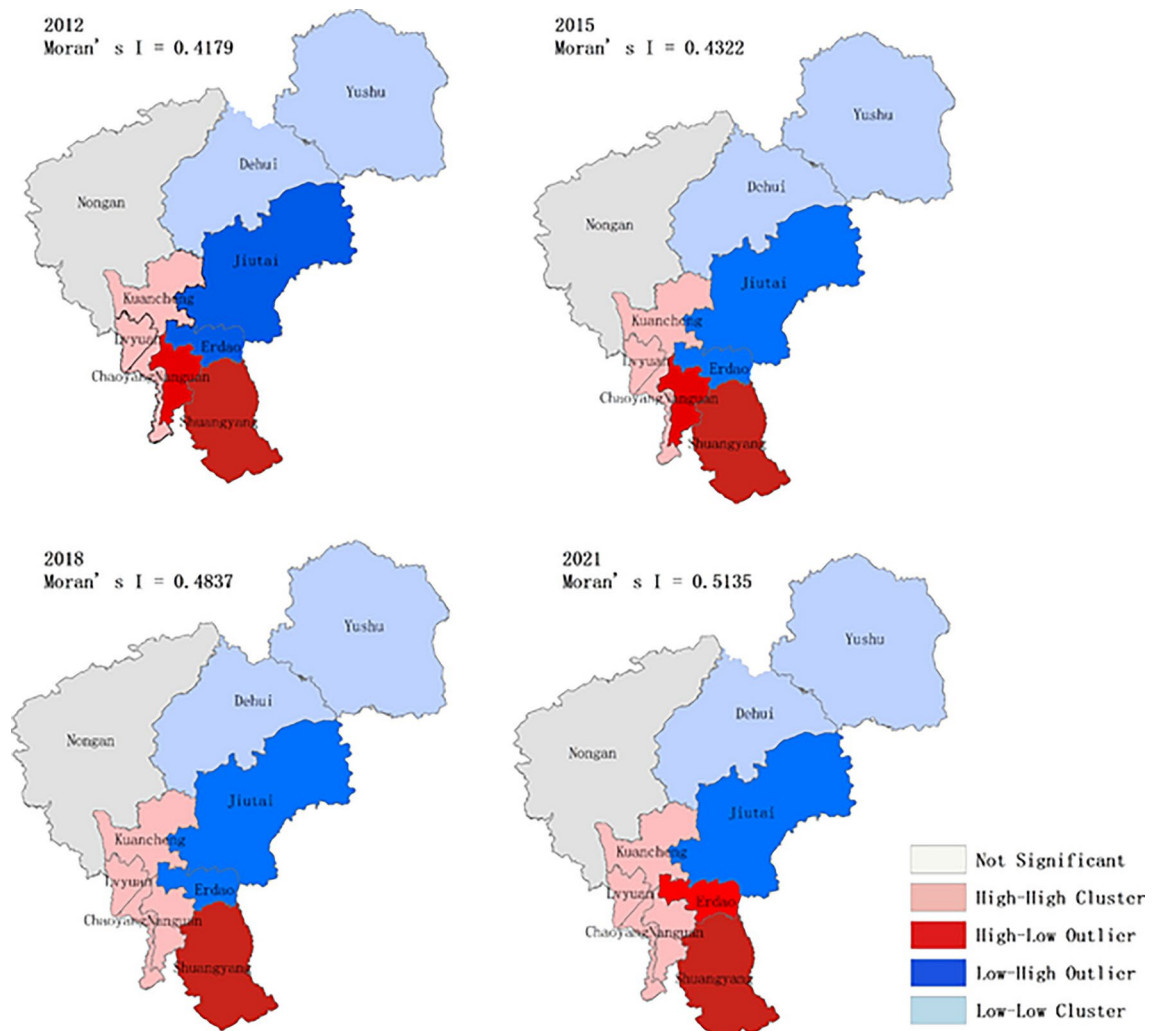


Fig. 11. Moran's I Statistics and Clusters, 2012 to 2021.

In this study, the improved AHP and entropy weight method were used to assign weights to each indicator, and the spatial resilience of the transport system was analyzed by combining GIS technology. This method improves the objectivity and accuracy of the assessment to a certain extent, but it still has limitations in some aspects. Firstly, the weight allocation, although effective, may need to be adjusted more carefully in different regions and climatic conditions. Future research can introduce a dynamic weight adjustment mechanism so that the assessment model can be flexibly adapted to the actual regional conditions. Second, the time span and precision of the data also impose some limitations on the predictive ability of the model. It is recommended that future research incorporate data from longer time series to enhance the stability and applicability of the assessment results.

Despite the proposed resilience enhancement strategies for urban–rural and remote areas, there are still many challenges to actual implementation. First, the shortage of funds and resources is the main obstacle to enhancing resilience in these regions. To address this issue, the government should increase financial support to these regions and introduce public–private partnership models to attract private capital to participate in infrastructure development. In addition, to address the problem of weak emergency management capacity, it is recommended to establish a cross-regional emergency linkage mechanism to optimize resource deployment and improve the systematic and efficient emergency response. At the policy level, transport resilience enhancement requires long-term planning. It is recommended that the government introduce special transport resilience construction standards and incorporate them into urban planning regulations to ensure that newly built and renovated transport facilities are sufficiently disaster-resistant. In addition, through the establishment of an intelligent disaster early warning system and emergency management platform, real-time monitoring and scheduling of the transport system can be realized to enhance the response efficiency in the event of a disaster.

The multidimensional resilience assessment framework of this study provides an important reference for the performance of the transport system in Changchun under extreme weather. Future studies should consider the actual situation of different cities to further validate and improve the model to enhance the resilience and recovery of the transport system.

Conclusion

The multidimensional resilience assessment framework developed in this study provides a scientific and systematic approach to quantitatively assess the performance of urban transport systems in natural disasters. Through empirical analyses of Changchun City's transport system, the study has produced the following key results:

1. application of the comprehensive assessment framework: this study successfully constructed a multi-dimensional assessment framework based on natural disasters, infrastructure vulnerability, social exposure and emergency response capacity, which can comprehensively reveal the performance of the urban transport system under extreme climate.
2. the discovery of regional resilience differences: the study shows that Changchun city center area is more resilient, while the urban–rural and peripheral areas are less resilient, and priority should be given to improving the infrastructure construction and emergency management capacity of low resilience areas in the future.
3. Scientific nature of quantitative assessment method: Through the improved AHP method and entropy weighting method, this study ensures the scientific nature and objectivity of the weight allocation in the multi-dimensional assessment process, which provides a reference and methodological support for future transport resilience research.
4. Suggestions for differentiated strategies: In view of the differences in the resilience of different regions in Changchun, strategies are proposed to strengthen the infrastructure and emergency resource allocation in the peripheral regions, so as to achieve the balanced development and sustainability of the urban transport system.

This study not only provides a scientific basis for the enhancement of transport resilience in Changchun, but also provides a referable theoretical framework and practical strategies for other cities in coping with extreme weather and disaster management. Future studies can be further expanded to more cities to verify the broad applicability of this assessment framework.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Received: 13 May 2024; Accepted: 25 December 2024

Published online: 07 January 2025

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Conceptualization, Z.W.; Data curation, Z.W. and G.L.; Formal analysis, Z.W. and J.Z.; Methodology, Z.W. and Q.L.; Writing—Original draft, Z.W.; Funding acquisition, Y.Z. and H.L.; Writing—Review and editing, J.Z. and Y.C. All authors have read and agreed to the published version of the manuscript.

Funding

This research was funded by the Key Scientific and Technology Research and Development Program of Jilin Province (20200403074SF); the Key Scientific and Technology Research and Development Program of Jilin Province (20180201033SF); the Key Scientific and Technology Research and Development Program of Jilin Province (20180201035SF); the Key Science and Technology Development Program Research and Development Projects of Jilin Province (20220203187SF).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-84672-x>.

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